

Machine Learning

Revision Pack: Lectures 1 to 4

Revision sheets, understanding questions, quizzes and maths questions for every chapter, with full answers.
Numerical applications are collected at the end.

How to use this pack

Each chapter has four parts.

- The **revision sheet** (teal box) lists every concept and formula of the lecture on about one page. Read it first.
- The **understanding questions** (U) ask you to explain in words. Answer each one aloud or on paper before reading the green answer under it.
- The **quiz** has true/false and multiple-choice questions; the answer key with one-line reasons follows the quiz.
- The **maths questions** (M) are derivations and proofs, not computations: definitions, expansions, inequalities and limits. Difficulty is marked ★ (direct), ★★ (several steps) or ★★★ (full proof).

Numbers appear only in the last section, *Numerical applications*, which applies the formulas of all four chapters once. Notation follows the one-read document of Lectures 1 to 4, and the maths questions avoid repeating the earlier maths problem sets.

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1 Lecture 1: What does it mean to learn?

Revision sheet: Lecture 1

Learning. Capacity to learn + experience $\xrightarrow{\text{learning}}$ acquired abilities. The abilities: *recognise* (what is it?), *predict* (what may happen?), *reason* (what follows?), *act* (what should I do?).

Two ways to give a machine an ability.

- *Encode human knowledge as rules.* This is right whenever a reliable procedure is known: sorting, tax computation, password checking. Expert systems (MYCIN) hit the **knowledge acquisition bottleneck**, which has three separate parts: obtaining knowledge from experts, representing it as rules, and facing situations the expert never saw.
- *Make the machine learn from experience.* This is the right choice when an ability exists but has no explicit procedure: *having an ability* $\neq$ *having an explicit procedure*. Machine learning is still programming: the designer decides what can be represented, what can change, what guides the change and what counts as good.

Three forms of experience, three settings.

Experience	Content	Setting
examples	input + desired output	supervised
observations	raw data, no answers	unsupervised / self-supervised
interactions	actions + consequences (rewards)	reinforcement learning

Supervised learning. Labels (targets, ground truth) come from measurement, annotation, or the future, and they can be noisy, disputed or convention-dependent: label quality bounds what can be learned. Training cycle: predict $\rightarrow$ compare with label $\rightarrow$ update. The *output* decides the problem type: numerical output is **regression**, categorical output is **classification**.

Prediction / decision / action. What the model says / the response chosen with a criterion / what is executed. They differ exactly when different errors have different costs.

Representation. Feature engineering means humans choose what is presented to the model. Representation learning means the representation is learned too, which deep layers suit. *Self-supervision* builds a signal from observations (predict a masked word, region or next token), so large unlabelled collections become useful and transferable. An autoregressive language model is self-supervised, and generation is repeated next-token prediction.

Four ingredients. Experience, model, learning algorithm, evaluation criterion.

Generalisation. Performing well on situations not seen in learning. Fit on seen examples is only partial evidence. Watch for spurious cues (wolves on snow), for diversity versus amount of data, and for the gap between simulation and reality.

Information theory. Entropy $H(p) = -\sum_c p_c \log p_c$ measures uncertainty. An input is useful when it carries information about the output, measured by the mutual information $I(X; Y) = H(Y) - H(Y | X)$. Cross-entropy measures the quality of predicted probabilities (Lecture 2).

1.1 Understanding questions

U1.1. Why is sorting a list a poor candidate for machine learning, while recognising handwritten digits is a good one?

Answer

Sorting has a known, reliable, exact procedure. Learning it from examples could only add errors and uncertainty, so encoding the rule is the right approach. Digit recognition is an ability people have but cannot write down as a procedure: rules about strokes and curves meet endless exceptions (position, size, slant, thickness). Examples of the ability are plentiful, so learning from them is attractive. The criterion is whether an explicit procedure exists, not how “intelligent” the task

looks.

U1.2. The “knowledge acquisition bottleneck” hides three different difficulties. Name them and give one example of each in a medical expert system.

Answer

(1) *Obtaining* the knowledge: experts cannot always say what they do (a radiologist “sees” a pattern without being able to state rules). (2) *Representing* it: even stated knowledge is hard to encode as consistent rules (interacting symptoms, exceptions, degrees of certainty). (3) *Coverage*: the system meets situations the expert never described (a new disease variant, an unusual combination of tests). These are different problems, and fixing one does not fix the others.

U1.3. A hospital (i) predicts pneumonia from labelled X-rays, (ii) groups unlabelled X-rays, (iii) trains a model to predict a removed image region, and (iv) trains a robot to position a sensor using a reward. Name the setting of each, and say what feedback tells the learner it is doing well.

Answer

(i) Supervised: the label (pneumonia or not) is the feedback. (ii) Unsupervised: there is no external feedback, only internal criteria such as compact groups. (iii) Self-supervised: the hidden region itself is the target, built from the observation. (iv) Reinforcement learning: the reward after actions. Here the learner’s actions also change what it experiences next.

U1.4. From the same weather data, state a regression task and a classification task. What decides which one you have?

Answer

Predicting the rainfall amount in mm is regression, and predicting whether it rains tomorrow is classification. The *output* decides, never the input: the same data serve both.

U1.5. A model outputs $P(\text{attack} \mid x) = 0.3$. Why is this not yet an alert? Describe two applications where the same 0.3 leads to different actions.

Answer

0.3 is a *prediction*: a summary of the evidence. An alert is a *decision*, and it also depends on what each error costs. In a critical infrastructure network, where a missed attack is catastrophic and an alarm is cheap, 0.3 should trigger an alert. In a consumer setting with alarm fatigue, where false alarms are costly and attacks minor, 0.3 should not. The probability is the same in both; only the loss differs (Lecture 2 makes this a threshold).

U1.6. What does feature engineering decide, and what is left to the model? What changes with representation learning, and what do humans still design?

Answer

With feature engineering, humans decide *what* is presented (edges, MFCCs, chosen house attributes), and the model learns *how* to use it. With representation learning, the features themselves are learned from data, typically layer by layer from local patterns to abstract properties. Humans still choose the architecture, the learning method, the data and the objective.

U1.7. Why is self-supervised learning useful when labelled data are scarce?

Answer

It turns abundant unlabelled observations into a learning signal: predict a missing part from the rest. Solving that task forces the model to capture regularities of the data. The resulting representation can be transferred and fine-tuned on the small labelled task, so far fewer labels are needed.

U1.8. A wolf-versus-dog classifier works in development and fails in deployment. Wolves were photographed on snow. How would you diagnose the problem, and which of the four ingredients would you change?

Answer

The model used the background (snow) instead of the animal: a spurious cue. Diagnose it with counter-examples, a wolf on grass and a dog on snow, or by looking at which image regions drive the decision. The fix is mainly the *experience*: collect images with the background varied independently of the class. The *evaluation criterion* should also include test cases that break the correlation.

U1.9. A million bird photos from one camera in one park, or fifty thousand from many people, countries, seasons and angles: which do you prefer, and why?

Answer

The diverse fifty thousand. Generalisation needs the training experience to cover the variation met in deployment. A million near-identical views add little information about new conditions. The amount of data is not the same as its diversity and representativeness.

U1.10. Why is the claim “our AI has learned every possible situation” incoherent?

Answer

Experience is finite and deployment is open-ended. The situations met after deployment are by definition not the ones observed. A system can only generalise from what it saw, under assumptions about how new situations resemble old ones. It cannot have seen all of them.

1.2 Quiz

- Q1.** Predicting a house price from a photo is classification, because the input is an image. [True / False]
- Q2.** Which form of experience gives direct guidance about the desired output?
(a) observations (b) examples (c) interactions (d) none of them
- Q3.** In reinforcement learning the learner partly shapes its future experience. [True / False]
- Q4.** The main practical limit of expert systems was
(a) processor speed (b) the knowledge acquisition bottleneck (c) storage (d) the impossibility of rules
- Q5.** Training an autoregressive language model is supervised learning with human labels. [True / False]
- Q6.** Which is *not* one of the four ingredients of a learning system?
(a) experience (b) model (c) GPU (d) evaluation criterion
- Q7.** A model that is perfect on its training examples has learned the task. [True / False]
- Q8.** Representation learning means
(a) humans design the features (b) the model learns useful features from data (c) no human choices remain (d) a method only for text
- Q9.** A decision depends on the prediction *and* on the consequences of errors. [True / False]
- Q10.** Grouping student trajectories with no target variable is
(a) regression (b) classification (c) unsupervised learning (d) reinforcement learning

Q11. Writing explicit rules remains the right approach when a reliable procedure is known. [True / False]

Q12. Entropy measures

- (a) accuracy (b) uncertainty (c) dataset size (d) training loss

Answer key

- Q1 False: the *output* (a price) makes it regression.
 Q2 (b) examples.
 Q3 True: its actions change what happens next.
 Q4 (b).
 Q5 False: it is self-supervised; the next token comes from the text itself.
 Q6 (c).
 Q7 False: training fit is only partial evidence; generalisation is the goal.
 Q8 (b).
 Q9 True.
 Q10 (c).
 Q11 True: learning would only add uncertainty.
 Q12 (b).

1.3 Maths questions

M1.1 (★★) Gibbs' inequality and the range of entropy

Let p, q be distributions on $\{1, \dots, C\}$, with $q_c > 0$ wherever $p_c > 0$. Use the inequality $\ln t \leq t - 1$ for $t > 0$, with equality only at $t = 1$.

(a) Prove $D_{KL}(p||q) = \sum_c p_c \ln \frac{p_c}{q_c} \geq 0$, with equality if and only if $p = q$.

Answer

Sum over $S = \{c : p_c > 0\}$ (terms with $p_c = 0$ contribute 0):

$$-D_{KL}(p||q) = \sum_{c \in S} p_c \ln \frac{q_c}{p_c} \leq \sum_{c \in S} p_c \left(\frac{q_c}{p_c} - 1 \right) = \sum_{c \in S} q_c - 1 \leq 1 - 1 = 0.$$

Equality requires $q_c/p_c = 1$ on S (equality in $\ln t \leq t - 1$) and $\sum_{c \in S} q_c = 1$. Together these give $q = p$. ■

(b) Deduce $0 \leq H(p) \leq \ln C$, and identify when each bound is attained.

Answer

Lower bound. Each term $-p_c \ln p_c \geq 0$ since $0 \leq p_c \leq 1$. The sum is 0 exactly when every $p_c \in \{0, 1\}$: a deterministic outcome.

Upper bound. With u the uniform distribution, $D_{KL}(p||u) = \sum_c p_c \ln(Cp_c) = \ln C - H(p) \geq 0$ by (a). Equality holds exactly when $p = u$. The uniform distribution is the most uncertain. ■

M1.2 (★★) Observing an input never increases uncertainty, on average

For discrete X, Y define $H(Y | X) = \sum_x p(x) H(Y | X = x)$ and $I(X; Y) = H(Y) - H(Y | X)$.

(a) Show that $I(X; Y) = D_{KL}(p_{XY} || p_X p_Y)$, and conclude $H(Y | X) \leq H(Y)$, with equality if and only if X and Y are independent.

Answer

Write both entropies as sums over the joint distribution, using $\sum_x p(x, y) = p(y)$:

$$H(Y) = - \sum_{x,y} p(x, y) \ln p(y), \quad H(Y | X) = - \sum_{x,y} p(x, y) \ln p(y | x).$$

Subtracting, $I(X; Y) = \sum_{x,y} p(x, y) \ln \frac{p(y|x)}{p(y)} = \sum_{x,y} p(x, y) \ln \frac{p(x,y)}{p(x)p(y)} = D_{KL}(p_{XY} \| p_X p_Y)$. By M1.1 this is ≥ 0 , with equality if and only if $p(x, y) = p(x)p(y)$ for all (x, y) , that is, independence. ■

(b) Show by an example that a *particular* observation can increase uncertainty: $H(Y | X = x) > H(Y)$ is possible.

Answer

Take $P(X = a) = 0.2$, $P(X = b) = 0.8$, with $Y | a$ uniform on $\{0, 1\}$ and $Y | b$ always 1. Then $P(Y = 1) = 0.2(0.5) + 0.8 = 0.9$, so $H(Y) = H_b(0.9) \approx 0.325$ nats. But $H(Y | X = a) = \ln 2 \approx 0.693 > 0.325$. On average, $H(Y | X) = 0.2 \ln 2 \approx 0.139 \leq 0.325$, as (a) requires. A rare, ambiguous input can leave you less sure than before; the average cannot.

M1.3 (★) The chain rule behind autoregressive models

A sequence $w_{1:T} = (w_1, \dots, w_T)$ and the convention $p(w_1 | w_{<1}) = p(w_1)$.

(a) Prove by induction that $p(w_{1:T}) = \prod_{t=1}^T p(w_t | w_{<t})$.

Answer

$T = 1$ is the convention. If the formula holds for T , the definition of conditional probability gives $p(w_{1:T+1}) = p(w_{T+1} | w_{1:T}) p(w_{1:T}) = p(w_{T+1} | w_{<T+1}) \prod_{t=1}^T p(w_t | w_{<t})$. No independence assumption is used. ■

(b) Deduce that the negative log-likelihood of a text is a sum of next-token cross-entropies. Why does this make next-token training equivalent to maximum likelihood on whole texts?

Answer

Take $-\log$ of (a): $-\log p_\theta(w_{1:T}) = \sum_{t=1}^T [-\log p_\theta(w_t | w_{<t})]$. Each term is the cross-entropy of the predicted next-token distribution against the one-hot target w_t . Minimising the sum of per-token losses is therefore exactly maximising the probability of the whole text. Sampling one token at a time from the factors draws a whole sequence from the joint distribution, which is why generation is repeated prediction.

M1.4 (★★) Memorising is not learning

X has a continuous distribution, so $P(X = x) = 0$ for every x . Given training pairs with distinct inputs, the memoriser outputs $f_{\text{mem}}(x) = y^{(i)}$ if $x = x^{(i)}$ for some i , and $d(x)$ (a fixed fallback rule) otherwise.

(a) Show that its training zero-one error is 0, but that its population risk equals that of the fallback: $R(f_{\text{mem}}) = R(d)$.

Answer

On each training input it returns the stored label, so the training error is 0. The two rules differ only on $S = \{x^{(1)}, \dots, x^{(N)}\}$, and $P(X \in S) \leq \sum_i P(X = x^{(i)}) = 0$. Splitting the expectation on

the events $X \in S$ and $X \notin S$:

$$R(f_{\text{mem}}) = \mathbb{E}[\ell(Y, f_{\text{mem}}(X))\mathbf{1}(X \notin S)] + 0 = \mathbb{E}[\ell(Y, d(X))\mathbf{1}(X \notin S)] = R(d). \quad \blacksquare$$

(b) What does this say about training error as evidence of learning?

Answer

Zero training error is compatible with having learned *nothing* beyond the fallback. Training error measures fit to examples that new inputs will (almost surely) never repeat. Only performance on new draws, the population risk estimated on held-out data, shows generalisation.

M1.5 (**) A representation can only lose information for decisions

An event Z is encoded as $X = g(Z)$ for a fixed function g . Let $E_X = \min_h P(h(X) \neq Y)$ and $E_Z = \min_k P(k(Z) \neq Y)$ be the smallest achievable error rates (Bayes errors).

(a) Prove $E_X \geq E_Z$.

Answer

For any classifier h of X , the composition $k = h \circ g$ is a classifier of Z with $P(k(Z) \neq Y) = P(h(X) \neq Y)$. So the set of error rates achievable from X is a subset of those achievable from Z , and a minimum over a subset is at least the minimum over the whole set. $\blacksquare$

(b) Deduce that adding a feature X' cannot increase the Bayes error. Then show that if X' is irrelevant, that is $P(Y | X, X') = P(Y | X)$, the Bayes error is unchanged.

Answer

$X = \pi(X, X')$ for the projection π , so (a) with $Z = (X, X')$ gives $E_X \geq E_{(X, X')}$. If X' is irrelevant, the posteriors coincide, so the Bayes classifier $\arg \max_c P(Y = c | X, X') = \arg \max_c P(Y = c | X)$ uses only X , and the two errors are equal. Irrelevant features cost nothing in the limit of infinite data, but they make *finite-sample* estimation harder (Lecture 2).

2 Lecture 2: From representations to decisions

Revision sheet: Lecture 2

Input as a vector. $x = (x_1, \dots, x_d)^\top$; categorical features are one-hot encoded. The representation fixes what information exists: two different events mapped to the same x create unavoidable (Bayes) error.

Decisions with a known distribution.

$$\text{conditional risk} \quad r(x, a) = \mathbb{E}[\ell_{\text{dec}}(Y, a) \mid X = x]$$

$$\text{Bayes rule} \quad f^*(x) \in \arg \min_a r(x, a)$$

Squared loss gives the conditional *mean*; absolute loss the *median*; pinball loss (c_u per unit under, c_o per unit over) the quantile $F(a^*) = \frac{c_u}{c_u + c_o}$. Zero-one loss gives $\arg \max_c P(Y = c \mid x)$.

With costs: decide positive iff $p > \frac{C_{FP}}{C_{FP} + C_{FN}}$. With a review action of cost C_R : benign for $p \leq C_R/C_{FN}$, attack for $p \geq 1 - C_R/C_{FP}$, review in between.

From P to data. The dataset $\mathcal{D} = \{(x^{(i)}, y^{(i)})\}_{i=1}^N$ is assumed i.i.d. from $P(X, Y)$.

KNN. $\hat{f}_k(x) = \frac{1}{k} \sum_{i \in N_k(x)} y^{(i)}$ and $\hat{P}_k(c \mid x) = \frac{1}{k} \sum_{i \in N_k(x)} \mathbf{1}(y^{(i)} = c)$, with a majority vote for classification. Distances: Minkowski d_p , weighted. Always standardise. k is a hyperparameter: small k means high capacity. Curse of dimensionality: a neighbourhood must grow to contain points, $P(M_d \geq 1) = 1 - (1 - r^d)^n$.

Parameterised models. $\mathcal{H} = \{f_\theta : \theta \in \Theta\}$. A classifier outputs $\hat{p}_\theta(x) \in \Delta^{C-1}$, and the decision rule is applied afterwards.

Cross-entropy. $\ell_{CE} = -\log \hat{p}_{\theta, y}(x)$, $\hat{R}_{CE} = -\frac{1}{N} \sum_i \log \hat{p}_{\theta, y^{(i)}}(x^{(i)})$, binary $\ell_{BCE} = -y \log \hat{p} - (1 - y) \log(1 - \hat{p})$. It is a smooth surrogate (zero-one loss has zero gradient almost everywhere). $H(p, \hat{p}) = H(p) + D_{KL}(p \parallel \hat{p})$ is minimised at $\hat{p} = p$. $R_{CE} = H(Y \mid X) + \mathbb{E}_X D_{KL}$.

ERM. $R(\theta) = \mathbb{E}[\ell(Y, f_\theta(X))]$; $\hat{R}_{\text{train}}(\theta) = \frac{1}{N} \sum_i \ell(y^{(i)}, f_\theta(x^{(i)}))$; $\hat{\theta} \in \arg \min_\theta \hat{R}_{\text{train}}$. For a fixed θ : $\mathbb{E} \hat{R} = R$, $\text{Var} \hat{R} = \sigma_\ell^2 / N$. ERM fits parameters but cannot choose complexity, and it is silent between functions that agree on the data.

Parameters versus hyperparameters (learned on train / chosen on validation). **Objective versus algorithm.** **Train/validation/test:** fit / select / measure once. K -fold CV: $\hat{R}_{CV}(\lambda) = \frac{1}{K} \sum_j \hat{R}_{\text{val}}^{(j)}(\lambda)$, then refit on all development data.

2.1 Understanding questions

U2.1. Why does “the best prediction” have no meaning until the loss is fixed? Illustrate with one conditional distribution.

Answer

Different losses penalise different errors, so they are minimised by different summaries of the same $P(Y \mid x)$. For a skewed distribution of repair times, squared loss picks the mean, absolute loss the median, and a loss where under-estimating is four times costlier picks the 80% quantile. These are three different numbers for one distribution. A model trained only to output the mean cannot serve the asymmetric loss later.

U2.2. Why is moving the decision threshold to reflect costs *not* a fix for class imbalance?

Answer

The cost threshold acts on a posterior that is assumed correct: it chooses the action given the evidence. Imbalance is an *estimation* issue: with few minority examples the posterior itself may be poorly estimated. If the posterior is wrong, changing the threshold does not repair it; if it is

right and reflects the true prevalence, imbalance is not an error at all. Two different questions.

U2.3. Two physically different network events, one benign and one malicious, map to the same feature vector. What follows for every classifier? Distinguish a legitimate new feature from leakage.

Answer

Any deterministic classifier outputs one label for that x , so it is wrong on one of the two events. The resulting error belongs to the problem as represented (Bayes error), not to the labels. A new feature that separates them is legitimate if it is available at decision time (the destination port), and leakage if it is only known later (an analyst's incident ticket).

U2.4. Why must features be standardised before KNN? What hidden assumption does “one-hot encoding plus Euclidean distance” make?

Answer

Distances add squared differences across features, so the feature with the largest units dominates the neighbourhood (income in euros swamps age in years). Standardising puts features on comparable scales, and the scaling is a modelling choice. With one-hot plus Euclidean distance, every pair of distinct categories is at the same distance $\sqrt{2}$, so “neighbouring” and “opposite” locations look equally different. A domain-aware distance avoids this.

U2.5. What does k control in KNN? Why is it a hyperparameter, and what happens at the extremes $k = 1$ and $k = N$?

Answer

k controls how locally the estimate averages: small k follows individual points (high capacity, noisy, jagged boundary), while large k smooths (low capacity, can erase real structure). It is not fitted by the averaging rule itself, so it is chosen on validation data. At $k = 1$ the training error is 0 (each point is its own neighbour). At $k = N$ every input gets the same prediction: the overall mean, or the majority class.

U2.6. Explain the curse of dimensionality in one paragraph, and say why it motivates parametric models.

Answer

Local estimation needs examples near the query. In high dimension, volume grows so fast that a fixed dataset is sparse. A neighbourhood containing even one point must cover a large fraction of the space, so “local” averages become global and lose their meaning. Parametric models avoid this by sharing structure: a few parameters describe the function everywhere and are fitted with *all* examples instead of the nearby few.

U2.7. Why is zero-one loss not used to train models? What does cross-entropy reward that zero-one loss ignores?

Answer

Zero-one loss is piecewise constant in the parameters: its gradient is zero almost everywhere and undefined where the predicted class flips, so gradient methods get no signal. It also treats a hesitant and a confident correct answer alike. Cross-entropy is smooth and graded: it rewards putting more probability on the true class and punishes confident mistakes heavily. It is the surrogate that learns good probabilities.

U2.8. Model A gives probability 0.6 to the true class on all four test cases. Model B gives 0.99 on three cases and 0.2 on one. Accuracy prefers A, but cross-entropy prefers B. Is this a contradiction?

Answer

No. Accuracy at threshold 0.5 judges the *decisions*; cross-entropy judges the *probabilities*. B is very informative on three cases and wrong on one, while A is uninformative everywhere. They answer different questions: the training loss is for estimating probabilities, and the decision loss is applied afterwards.

U2.9. Why is the training average an unbiased estimate of $R(\theta)$ for a fixed θ , but not of $R(\hat{\theta})$ for the ERM solution?

Answer

For a fixed θ , each $\ell_i(\theta)$ is an independent draw of the loss, so the average has mean $R(\theta)$. The ERM solution $\hat{\theta}$ is chosen *using* those same examples, to make that very average small. The examples then no longer behave like fresh draws for $\hat{\theta}$, and the average is biased downward (Lecture 3 proves it).

U2.10. What is each of the training, validation and test sets for? What goes wrong if the test set is consulted during development?

Answer

The training set fits the parameters and every data-dependent preprocessing step. The validation set compares configurations and chooses hyperparameters and thresholds. The test set measures once, after everything is frozen. Using test results for any choice turns the test set into another validation set: the selected model is adapted to it, and its reported score becomes optimistic.

U2.11. One team picks the polynomial degree on validation data; another always uses degree 20 with an L_1 penalty. Why is “degree” a hyperparameter in one system but not in the other?

Answer

Whether something is a parameter or a hyperparameter depends on the *procedure*, not on the quantity itself. In the first system the degree is chosen outside the fit, on validation data. In the second the degree is fixed, and the penalty strength is the hyperparameter; the effective degree emerges from the fit as some coefficients go to zero.

2.2 Quiz

Q1. Under absolute loss the Bayes predictor is the conditional

- (a) mean (b) median (c) mode (d) maximum

Q2. Under zero-one loss the Bayes classifier picks the most probable class. [True / False]

Q3. With $C_{FP} = 1$ and $C_{FN} = 4$, declare positive when p exceeds

- (a) 0.2 (b) 0.25 (c) 0.5 (d) 0.8

Q4. Raising C_{FN} raises the decision threshold. [True / False]

Q5. KNN classification with $k = N$ predicts

- (a) the nearest point's class (b) the majority class of the whole training set (c) a random class (d) the minority class

Q6. Smaller k means lower capacity. [True / False]

Q7. Standardising the features can change KNN predictions. [True / False]

Q8. The cross-entropy of putting probability 0.5 on the true class is

- (a) 0.5 (b) $\ln 2$ (c) 1 (d) 0

Q9. $H(p, q) = H(p) +$

- (a) $H(q)$ (b) $D_{KL}(p||q)$ (c) $D_{KL}(q||p)$ (d) $I(X;Y)$

Q10. For the ERM solution, $\mathbb{E}_{\mathcal{D}}[\hat{R}_{\text{train}}(\hat{\theta})] = R(\hat{\theta})$. [True / False]

Q11. Hyperparameters should be selected on

- (a) training data (b) validation data (c) test data (d) all data

Q12. In K -fold cross-validation each example is used for validation

- (a) K times (b) once (c) $K - 1$ times (d) never

Q13. The Bayes error can be strictly positive even with perfectly correct labels. [True / False]

Q14. For fixed θ , the variance of the training average is

- (a) σ_{ℓ}^2 (b) σ_{ℓ}^2/N (c) σ_{ℓ}/N (d) $\sigma_{\ell}^2/\sqrt{N}$

Q15. In “Bayes decision rule”, “Bayes” refers to a prior on the parameters. [True / False]

Answer key

- Q1 (b).
 Q2 True.
 Q3 (a): $1/(1+4)$.
 Q4 False: a costlier miss *lowers* the threshold.
 Q5 (b).
 Q6 False: smaller k means *higher* capacity.
 Q7 True: it changes the geometry and hence the neighbours.
 Q8 (b): $-\ln 0.5 \approx 0.693$.
 Q9 (b).
 Q10 False: it is optimistic.
 Q11 (b).
 Q12 (b).
 Q13 True: overlapping classes in the representation.
 Q14 (b).
 Q15 False: it means optimal decision under a probability distribution.

2.3 Maths questions

M2.1 (★) Why the pointwise rule is globally optimal

For any decision function f , let $R(f) = \mathbb{E}[\ell(Y, f(X))]$ and $r(x, a) = \mathbb{E}[\ell(Y, a) \mid X = x]$.

(a) Show that $R(f) = \mathbb{E}_X[r(X, f(X))]$, and deduce that the Bayes rule $f^*(x) \in \arg \min_a r(x, a)$ satisfies $R(f^*) \leq R(f)$ for every f .

Answer

By the tower property, $R(f) = \mathbb{E}_X[\mathbb{E}[\ell(Y, f(X)) \mid X]]$, and conditionally on $X = x$ the action $f(x)$ is a constant, so the inner term is $r(x, f(x))$. By definition of f^* , $r(x, f^*(x)) \leq r(x, f(x))$ for every x . Taking $\mathbb{E}_X$ of a pointwise inequality preserves it: $R(f^*) \leq R(f)$. Minimising the average is achieved by minimising *at every x separately*. ■

M2.2 (★★) Squared loss: the regression function and the excess risk

Let $m(x) = \mathbb{E}[Y \mid X = x]$.

(a) Prove $\mathbb{E}[(Y - a)^2 \mid X = x] = \text{Var}(Y \mid X = x) + (m(x) - a)^2$, and deduce that $f^* = m$, with Bayes risk $\mathbb{E}[\text{Var}(Y \mid X)]$.

Answer

Write $Y - a = (Y - m) + (m - a)$ and square. The cross term is $2(m - a) \mathbb{E}[Y - m \mid X = x] = 0$. This leaves the conditional variance, which does not depend on a , plus $(m - a)^2$, which is minimised

at $a = m(x)$. At the optimum only $\text{Var}(Y | x)$ remains, and averaging over X gives the Bayes risk $\mathbb{E}[\text{Var}(Y | X)]$. ■

(b) Show that the excess risk of any f is $R(f) - R(f^*) = \mathbb{E}[(f(X) - m(X))^2]$.

Answer

Apply (a) with $a = f(x)$ and average over X : $R(f) = \mathbb{E}[\text{Var}(Y | X)] + \mathbb{E}[(m(X) - f(X))^2]$. Subtract $R(f^*) = \mathbb{E}[\text{Var}(Y | X)]$. Learning a regressor under squared loss is learning to approximate m in mean square. ■

M2.3 (★★) The pinball loss gives a quantile

$Y | x$ has a continuous CDF F with density f . The loss is $\ell(y, a) = c_u(y - a)_+ + c_o(a - y)_+$, where $(t)_+ = \max(t, 0)$ and $c_u, c_o > 0$.

(a) Prove the identities $\mathbb{E}[(Y - a)_+] = \int_a^\infty (1 - F(t)) dt$ and $\mathbb{E}[(a - Y)_+] = \int_{-\infty}^a F(t) dt$. (Assume $\mathbb{E}|Y| < \infty$.)

Answer

For $u \geq 0$, $u = \int_0^\infty \mathbf{1}(u > s) ds$. With $u = (Y - a)_+$ and the substitution $t = a + s$: $(Y - a)_+ = \int_a^\infty \mathbf{1}(Y > t) dt$. Take expectations and swap them with the integral (the integrand is nonnegative, so this is allowed): $\mathbb{E}[(Y - a)_+] = \int_a^\infty P(Y > t) dt = \int_a^\infty (1 - F(t)) dt$. Symmetrically, $(a - Y)_+ = \int_{-\infty}^a \mathbf{1}(Y < t) dt$, whose expectation is $\int_{-\infty}^a F(t) dt$. ■

(b) Deduce that the expected loss $L(a)$ is convex and minimised where $F(a^*) = \frac{c_u}{c_u + c_o}$. Recover the median as a special case.

Answer

By (a) and the fundamental theorem of calculus,

$$L'(a) = -c_u(1 - F(a)) + c_o F(a), \quad L''(a) = (c_u + c_o)f(a) \geq 0.$$

So L is convex, and its minimum is where $L'(a) = 0$: $(c_u + c_o)F(a) = c_u$. With $c_u = c_o$ the level is $\frac{1}{2}$, the median. A costlier under-prediction ($c_u > c_o$) pushes the answer above the median. ■

M2.4 (★) Only cost differences matter

Binary decision with $p = P(Y = 1 | x)$ and a full cost matrix: C_{TP}, C_{FN} when $Y = 1$, and C_{FP}, C_{TN} when $Y = 0$, with $C_{FN} > C_{TP}$ and $C_{FP} > C_{TN}$.

(a) Show that the Bayes rule declares 1 if and only if $p > \frac{C_{FP} - C_{TN}}{(C_{FP} - C_{TN}) + (C_{FN} - C_{TP})}$.

Answer

The risk of declaring 1 is $pC_{TP} + (1-p)C_{FP}$, and of declaring 0 is $pC_{FN} + (1-p)C_{TN}$. Declaring 1 is better when $p(C_{FN} - C_{TP}) > (1-p)(C_{FP} - C_{TN})$. Rearranging, $p[(C_{FN} - C_{TP}) + (C_{FP} - C_{TN})] > C_{FP} - C_{TN}$, which is the stated condition. ■

(b) Explain why adding the same constant to both costs in one column (one true class) never changes the decision, and recover the lecture's threshold $C_{FP}/(C_{FP} + C_{FN})$.

Answer

The threshold depends only on the differences $C_{FP} - C_{TN}$ and $C_{FN} - C_{TP}$, called regrets. Adding a constant to both costs of the same true class leaves its difference unchanged. With free correct decisions, $C_{TP} = C_{TN} = 0$, which gives $C_{FP}/(C_{FP} + C_{FN})$.

M2.5 (★★) Log loss is strictly proper; zero-one loss is not

A binary outcome with $P(Y = 1) = p \in (0, 1)$, and a reported probability $q \in (0, 1)$.

(a) Show that the expected log loss $L(q) = -p \ln q - (1 - p) \ln(1 - q)$ has the unique minimiser $q = p$, with minimum value the binary entropy $H_b(p)$.

Answer

$L'(q) = -\frac{p}{q} + \frac{1-p}{1-q}$, which is zero if and only if $p(1 - q) = (1 - p)q$, that is $q = p$. Also $L''(q) = \frac{p}{q^2} + \frac{1-p}{(1-q)^2} > 0$, so this is the unique minimum, with $L(p) = -p \ln p - (1 - p) \ln(1 - p) = H_b(p)$. Equivalently, $L(q) - H_b(p) = D_{KL}((p, 1 - p) \parallel (q, 1 - q)) \geq 0$ (M1.1). ■

(b) Now score the report by zero-one loss after thresholding at $\frac{1}{2}$. Show that every $q > \frac{1}{2}$ has the same expected loss when $p > \frac{1}{2}$. Why does this make zero-one loss useless for learning probabilities?

Answer

Any $q > \frac{1}{2}$ predicts class 1, with expected loss $1 - p$, whether $q = 0.51$ or $q = 0.99$. The loss cannot tell an accurate probability from a wildly overconfident one, so it gives no incentive to report $q = p$. A loss with $q = p$ as its *unique* optimum (strictly proper) is what makes cross-entropy learn calibrated probabilities.

M2.6 (★) ERM cannot decide between interpolating functions

Distinct points $x_1, \dots, x_N \in \mathbb{R}$, and a polynomial f with $f(x_i) = y_i$ for all i .

(a) Show that $f_\alpha(x) = f(x) + \alpha \prod_{i=1}^N (x - x_i)$ also interpolates the data for every $\alpha \in \mathbb{R}$, and that at any $x_0 \notin \{x_i\}$ the values $f_\alpha(x_0)$ cover all of $\mathbb{R}$ as α varies.

Answer

At each x_i the product contains the factor $(x_i - x_i) = 0$, so $f_\alpha(x_i) = f(x_i) = y_i$. At x_0 the product $\prod_i (x_0 - x_i)$ is a nonzero constant κ , so $f_\alpha(x_0) = f(x_0) + \alpha \kappa$ takes every real value. Infinitely many functions have zero training error and disagree arbitrarily off the data. ■

(b) What must break the tie?

Answer

Assumptions or new evidence: restricting the hypothesis space (a low degree), a penalty (regularisation), the algorithm's implicit preference, or validation data from the regions where the candidates disagree. This is inductive bias; the data alone cannot decide.

M2.7 (★★) Accuracy of the empirical risk for a fixed model

A fixed θ , i.i.d. examples, losses ℓ_i with mean R and variance σ_ℓ^2 . Chebyshev's inequality: $P(|Z - \mathbb{E}Z| \geq t) \leq \text{Var}(Z)/t^2$.

(a) Prove $\mathbb{E}\hat{R} = R$ and $\text{Var } \hat{R} = \sigma_\ell^2/N$, naming the assumption used at each step.

Answer

Mean. By linearity, $\mathbb{E}\hat{R} = \frac{1}{N} \sum_i \mathbb{E}\ell_i$. Each ℓ_i has mean R because each example is distributed as P (identical distribution) and θ does not depend on the data. So $\mathbb{E}\hat{R} = R$.

Variance. By independence the covariances vanish, so $\text{Var } \hat{R} = \frac{1}{N^2} \sum_i \text{Var } \ell_i = \sigma_\ell^2/N$. ■

(b) For zero-one loss, show that $N \geq \frac{1}{4\delta t^2}$ examples guarantee $P(|\hat{R} - R| \geq t) \leq \delta$, whatever R is.

Answer

Each ℓ_i is Bernoulli(R), so $\sigma_\ell^2 = R(1-R) \leq \frac{1}{4}$, the maximum of $u(1-u)$. Chebyshev gives $P(|\hat{R} - R| \geq t) \leq \frac{R(1-R)}{Nt^2} \leq \frac{1}{4Nt^2}$, which is $\leq \delta$ as soon as $N \geq 1/(4\delta t^2)$. ■

M2.8 (★) The 1-NN boundary between two points

Two training points $a \neq b$ in $\mathbb{R}^d$ with Euclidean distance.

(a) Show that $\{x : \|x - a\| = \|x - b\|\}$ is the hyperplane $2(b-a)^\top x = \|b\|^2 - \|a\|^2$, perpendicular to $b - a$ and passing through the midpoint $\frac{a+b}{2}$.

Answer

Square both sides: $\|x\|^2 - 2a^\top x + \|a\|^2 = \|x\|^2 - 2b^\top x + \|b\|^2$. The $\|x\|^2$ terms cancel, leaving $2(b-a)^\top x = \|b\|^2 - \|a\|^2$: linear in x , with normal vector $b - a$. At $x = \frac{a+b}{2}$ the left side is $(b-a)^\top(a+b) = \|b\|^2 - \|a\|^2$, so the midpoint lies on it. ■

M2.9 (★★) How large must a neighbourhood be?

n points uniform in the unit ball of $\mathbb{R}^d$, a query at the centre, and M_d the number of points within radius $r < 1$.

(a) Show that $M_d \sim \text{Binomial}(n, r^d)$ and $P(M_d \geq 1) = 1 - (1 - r^d)^n$.

Answer

The ball of radius r has volume r^d times that of the unit ball, so each point falls inside independently with probability r^d . A count of independent successes is binomial, and $P(M_d \geq 1) = 1 - P(M_d = 0) = 1 - (1 - r^d)^n$. ■

(b) Find the radius r_d that gives $P(M_d \geq 1) = 1 - \delta$, and show that $r_d \rightarrow 1$ as $d \rightarrow \infty$ for fixed n and δ .

Answer

Solve $(1 - r^d)^n = \delta$: $r_d = (1 - \delta^{1/n})^{1/d}$. The base $\beta = 1 - \delta^{1/n}$ lies in $(0, 1)$ and does not depend on d , so $r_d = \beta^{1/d} = e^{(\ln \beta)/d} \rightarrow e^0 = 1$. To keep even one neighbour, the “neighbourhood” must grow to the whole ball: locality is lost. ■

3 Lecture 3: From fitting to generalisation

Revision sheet: Lecture 3

Two risks and the gap. $\hat{R}_{\mathcal{D}}(f) = \frac{1}{N} \sum_i \ell(y^{(i)}, f(x^{(i)}))$, $R(f) = \mathbb{E}[\ell(Y, f(X))]$, $G(\hat{f}_{\mathcal{D}}) = R(\hat{f}_{\mathcal{D}}) - \hat{R}_{\mathcal{D}}(\hat{f}_{\mathcal{D}})$. Training data are used twice, to select and to score, so $\mathbb{E}[\hat{R}_{\mathcal{D}}(\hat{\theta}_{\mathcal{D}})] \leq R(\theta_{\mathcal{H}}^*) \leq \mathbb{E}[R(\hat{\theta}_{\mathcal{D}})]$ and $\mathbb{E}G \geq 0$. Held-out data are unbiased for a frozen model; the winner of a selection is optimistic again (two equal models with noise σ : $R - \sigma/\sqrt{\pi}$).

Shortcut versus leakage. A shortcut is available at decision time but not a stable cause (background, time of day). Leakage is unavailable at decision time (a later ticket, test statistics).

Inductive bias enters through the data and representation, the hypothesis space, regularisation, and the optimiser.

Capacity. How varied $\mathcal{H}$ is; it is not the parameter count. Nested families give a monotone training error. Underfitting: both errors high. Overfitting: training low, validation high. Capacity is judged relative to N .

Bias–variance (squared loss, $f^* = \mathbb{E}[Y | x]$, $\bar{f} = \mathbb{E}_{\mathcal{D}} \hat{f}_{\mathcal{D}}$):

$$\mathbb{E}[(Y - \hat{f}_{\mathcal{D}}(x))^2] = \sigma^2(x) + (\bar{f}(x) - f^*(x))^2 + \mathbb{E}_{\mathcal{D}}[(\hat{f}_{\mathcal{D}}(x) - \bar{f}(x))^2].$$

Double descent: a peak at the interpolation threshold, then possibly a second descent; it is examined, not assumed. Excess error splits into approximation, estimation and optimisation error.

Controlling complexity. Regularised ERM: $\arg \min_{\theta} [\hat{R}_{\text{train}} + \lambda \Omega]$. L_2 shrinks every coefficient; L_1 sets some exactly to zero (the corners of the diamond). Larger λ means a smaller effective family. Early stopping: $t^* = \arg \min_t \hat{R}_{\text{val}}(\hat{\theta}_t)$, with patience. Feature selection: filters, forward and backward search, always redone inside each fold.

Evaluation. The split must reproduce the deployment novelty: random, stratified, grouped (by episode or device), or temporal. Metrics: accuracy, recall $\frac{TP}{TP+FN}$, precision $\frac{TP}{TP+FP}$, specificity, FPR, balanced accuracy, F_1 . Leakage takes many forms; fit preprocessing on training data only. Learning curves diagnose underfitting, variance, data limits and drift.

3.1 Understanding questions

U3.1. Why is the training error of the fitted model an optimistic estimate of its risk, even with i.i.d. data?

Answer

The same data choose the model and then score it. ERM picks the parameters that happen to fit *this* sample best, including its accidental regularities, so the score is biased downward. On average, the fitted model's training risk is below even the best achievable risk in the family, and its true risk is above that best risk.

U3.2. When does a test set give an unbiased estimate, and what destroys that property?

Answer

When the model is frozen before the test set is used, and the test examples are independent of everything that shaped the model. Then the model is fixed relative to them, and the average loss is unbiased. Any use of test results to make a choice (a configuration, a threshold, a feature) breaks this: the test set becomes a validation set, and the selected score is optimistic.

U3.3. Night-time flows are mostly attacks in the training data. Is “hour of day” leakage? What is it, and how would you protect against it?

Answer

It is not leakage: the hour is known at decision time. It is a *shortcut*, a cue that is predictive in the collected data but not a stable cause of attacks, and it can fail when attackers change schedule. Protect against it by collecting data where the nuisance factor varies, by evaluating on cases that break the correlation (daytime attacks), and by checking which features drive the predictions.

U3.4. List the four ways inductive bias enters a learning system, with one example of each.

Answer

(1) Data and representation: which features exist, such as images only, without metadata. (2) The hypothesis space: linear models cannot express curved boundaries. (3) Regularisation: L_2 prefers small weights. (4) The optimiser: gradient descent started from zero tends toward particular solutions among many minimisers.

U3.5. Why is capacity not the same as the number of parameters? Give two examples.

Answer

Capacity is how varied the functions the procedure can produce are. KNN has no trained parameters, yet with $k = 1$ it can fit any labelling. A large network with a strong penalty or early stopping has many parameters but a small effective family. Structure, constraints, regularisation, optimisation and the amount of data all enter.

U3.6. In plain words: what are bias and variance, what reduces each, and why does more data not remove bias?

Answer

Bias: the average fitted model, over many training sets, misses the target in a consistent direction. It is reduced by a richer family or better features. *Variance*: the fitted model changes when the training sample changes. It is reduced by more data, regularisation or lower capacity. More data from the same family only makes the fit converge to the best member of the family; if that member cannot express the truth, the gap (approximation bias) stays.

U3.7. Describe double descent. Why does the lecture insist that validation behaviour must be examined rather than assumed?

Answer

As capacity grows, validation error follows the classical U, peaks near the interpolation threshold (where the model can just fit every training point, noise included), and may fall again beyond it. In that region many interpolating solutions exist, and which one is selected (by the regulariser, the optimiser or the data geometry) decides the error. The same can happen over training time. Since none of this is guaranteed, the classical U is only a first model: you must look at the validation curve.

U3.8. Explain geometrically why L_1 produces exact zeros and L_2 does not.

Answer

The penalised solution is where the level sets of the training risk first touch the constraint region. The L_2 ball is round, so the contact point generically has every coordinate nonzero: coefficients shrink. The L_1 ball is a diamond with corners on the axes, and expanding ellipses usually hit a corner first, where some coordinates are exactly zero: the result is sparse.

U3.9. Why is early stopping a form of regularisation? What does patience trade off?

Answer

Training longer lets the model fit finer details, eventually noise. Stopping at the best validation checkpoint limits how far the parameters move, which reduces effective capacity at a fixed parameter count. Patience is how many epochs without improvement to wait. Too short stops on a noise fluctuation or before a second descent; too long costs computation, but the best checkpoint is kept anyway.

U3.10. Why must feature selection and standardisation be fitted inside each cross-validation fold?

Answer

They are data-dependent steps, hence part of the fitted model. If they are computed on all the data, information from each validation fold has already shaped the features or the scaling, and the validation score is no longer an honest estimate for unseen data. The fix is to fit them on the training folds only and apply them unchanged to the held-out fold.

U3.11. Which split do you use to estimate performance on (i) new independent cases, (ii) future traffic, and (iii) unseen devices or attack episodes?

Answer

(i) A random split, stratified if a class is rare. (ii) A temporal split: train on the past, test on later data. (iii) A grouped split, where each device or episode lies entirely in one subset. The split must reproduce the kind of novelty met in deployment.

U3.12. Attacks are 1% of flows. Why is accuracy misleading? What would you report instead?

Answer

“Always benign” scores 99% accuracy and catches nothing. Accuracy is dominated by the majority class. Report recall (how many attacks are caught), precision (how many alarms are real), the false-positive rate or alarms per volume of benign traffic, and a cost-weighted criterion. Also remember that recall on 100 attacks is noisy (standard error about 0.046 at 0.7).

3.2 Quiz

Q1. Enlarging a nested hypothesis family can increase the minimal training error. [True / False]

Q2. The expected generalisation gap of ERM is
(a) ≤ 0 (b) ≥ 0 (c) $= 0$ (d) undefined

Q3. The overfitting signature is
(a) both errors high (b) training low, validation high (c) both errors low (d) validation lower than training

Q4. More data from the same distribution removes bias when the family cannot express f^* . [True / False]

Q5. Which penalty can produce coefficients exactly equal to zero?
(a) L_2 (b) L_1 (c) both (d) neither

Q6. Increasing λ in ridge regression
(a) increases capacity (b) decreases effective capacity (c) has no effect (d) removes parameters

Q7. The noise term $\sigma^2(x)$ vanishes for the optimal predictor f^* . [True / False]

Q8. Two equal models, validation noise $\mathcal{N}(0, \sigma^2)$: the expected minimum validation error is
(a) R (b) $R - \sigma/\sqrt{\pi}$ (c) $R - \sigma$ (d) $R + \sigma/\sqrt{\pi}$

- Q9.** The standard scaler should be fitted on
 (a) the whole dataset (b) the training fold only (c) the test set (d) the validation set
- Q10.** To detect drift over time, use a
 (a) random split (b) temporal split (c) stratified split (d) bootstrap
- Q11.** Stratified splitting removes group leakage. [True / False]
- Q12.** The F_1 score ignores
 (a) TP (b) FP (c) TN (d) FN
- Q13.** A patience that is too short risks
 (a) leakage (b) stopping on a noise fluctuation (c) underfitting the test set (d) more parameters
- Q14.** Double descent happens in every over-parameterised model. [True / False]
- Q15.** Recall is computed over
 (a) all examples (b) actual positives (c) predicted positives (d) actual negatives

Answer key

- Q1 False: a minimum over a superset cannot be larger.
 Q2 (b).
 Q3 (b).
 Q4 False: it removes variance, not approximation bias.
 Q5 (b).
 Q6 (b).
 Q7 False: it is the irreducible part, present even for f^* .
 Q8 (b).
 Q9 (b).
 Q10 (b).
 Q11 False: it keeps class ratios; only grouping removes group leakage.
 Q12 (c).
 Q13 (b).
 Q14 False: it can happen, but it is not guaranteed.
 Q15 (b).

3.3 Maths questions

M3.1 (**) ERM is optimistic

$\theta_{\mathcal{H}}^* \in \arg \min_{\theta} R(\theta)$ (it does not depend on the data) and $\hat{\theta}_{\mathcal{D}} \in \arg \min_{\theta} \hat{R}_{\mathcal{D}}(\theta)$.

(a) Prove $\mathbb{E}_{\mathcal{D}}[\hat{R}_{\mathcal{D}}(\hat{\theta}_{\mathcal{D}})] \leq R(\theta_{\mathcal{H}}^*) \leq \mathbb{E}_{\mathcal{D}}[R(\hat{\theta}_{\mathcal{D}})]$, and conclude that $\mathbb{E}_{\mathcal{D}}[G] \geq 0$.

Answer

Left inequality. For every sample, $\hat{R}_{\mathcal{D}}(\hat{\theta}_{\mathcal{D}}) \leq \hat{R}_{\mathcal{D}}(\theta_{\mathcal{H}}^*)$, because $\hat{\theta}_{\mathcal{D}}$ minimises $\hat{R}_{\mathcal{D}}$. Take $\mathbb{E}_{\mathcal{D}}$. Since $\theta_{\mathcal{H}}^*$ is fixed, $\mathbb{E}_{\mathcal{D}}\hat{R}_{\mathcal{D}}(\theta_{\mathcal{H}}^*) = R(\theta_{\mathcal{H}}^*)$ (M2.7).

Right inequality. For every sample, $R(\hat{\theta}_{\mathcal{D}}) \geq R(\theta_{\mathcal{H}}^*)$, because $\theta_{\mathcal{H}}^*$ minimises R . Take $\mathbb{E}_{\mathcal{D}}$.

Gap. $\mathbb{E}G = \mathbb{E}R(\hat{\theta}_{\mathcal{D}}) - \mathbb{E}\hat{R}_{\mathcal{D}}(\hat{\theta}_{\mathcal{D}})$, the difference of the outer terms of the chain, which is ≥ 0 . ■

M3.2 (***) The bias–variance decomposition

$Y = f^*(X) + \varepsilon$ with $\mathbb{E}[\varepsilon | X] = 0$ and $\text{Var}(\varepsilon | X = x) = \sigma^2(x)$. The new noise ε is independent of the training sample $\mathcal{D}$, and $\bar{f}(x) = \mathbb{E}_{\mathcal{D}}\hat{f}_{\mathcal{D}}(x)$.

(a) Prove $\mathbb{E}_{\mathcal{D},\varepsilon}[(Y - \hat{f}_{\mathcal{D}}(x))^2] = \sigma^2(x) + (\bar{f}(x) - f^*(x))^2 + \mathbb{E}_{\mathcal{D}}[(\hat{f}_{\mathcal{D}}(x) - \bar{f}(x))^2]$, justifying why each cross term vanishes.

Answer

Write $Y - \hat{f}_D = A + B + C$ with $A = \varepsilon$, $B = f^* - \bar{f}$ (a constant), $C = \bar{f} - \hat{f}_D$, all at the fixed x . Expanding,

$$\mathbb{E}(A + B + C)^2 = \mathbb{E}A^2 + B^2 + \mathbb{E}C^2 + 2B\mathbb{E}A + 2\mathbb{E}[AC] + 2B\mathbb{E}C.$$

$\mathbb{E}A = 0$ by the noise assumption. $\mathbb{E}[AC] = \mathbb{E}[\varepsilon]\mathbb{E}[C] = 0$ because ε is independent of $\mathcal{D}$. $\mathbb{E}C = \bar{f} - \mathbb{E}_D \hat{f}_D = 0$ by the definition of $\bar{f}$. What remains is $\sigma^2(x) + \text{Bias}(x)^2 + \text{Var}(x)$. ■

M3.3 () How k trades bias for variance**

Fixed inputs $x_1, \dots, x_N$ (fixed design), $y_i = f^*(x_i) + \varepsilon_i$ with independent noise of mean 0 and variance σ^2 . KNN regression at a query x_0 averages the k nearest y_i , whose indices $N_k(x_0)$ are fixed because the inputs are fixed.

(a) Show that $\text{Bias}(x_0) = \frac{1}{k} \sum_{i \in N_k(x_0)} f^*(x_i) - f^*(x_0)$ and $\text{Var}(x_0) = \sigma^2/k$.

Answer

$\hat{f}_k(x_0) = \frac{1}{k} \sum_{i \in N_k} (f^*(x_i) + \varepsilon_i)$. Its mean is $\frac{1}{k} \sum_{i \in N_k} f^*(x_i)$, because the noise has mean 0; subtract $f^*(x_0)$ to get the bias. Its variance is that of $\frac{1}{k} \sum_{i \in N_k} \varepsilon_i$, which is $\frac{1}{k^2} \cdot k\sigma^2 = \sigma^2/k$ by independence. ■

(b) Explain how the two terms move when k increases, and why $k = 1$ is rarely optimal.

Answer

The variance σ^2/k falls as k grows. The bias usually grows, because farther neighbours have $f^*(x_i)$ further from $f^*(x_0)$ (unless f^* is flat). At $k = 1$ the variance is the full σ^2 , the noise of a single observation. A moderate k balances the two, and validation picks it.

M3.4 () Ridge regression has a unique closed-form solution**

$X \in \mathbb{R}^{N \times d}$, $y \in \mathbb{R}^N$, $\lambda > 0$, and $J(w) = \|y - Xw\|^2 + \lambda\|w\|^2$.

(a) Show that the minimiser satisfies $(X^\top X + \lambda I)w = X^\top y$, and that $X^\top X + \lambda I$ is invertible even when $d > N$.

Answer

$\nabla J(w) = -2X^\top(y - Xw) + 2\lambda w$. Setting it to zero gives $(X^\top X + \lambda I)w = X^\top y$. For $v \neq 0$, $v^\top(X^\top X + \lambda I)v = \|Xv\|^2 + \lambda\|v\|^2 > 0$. So the matrix is positive definite, hence invertible, whatever N and d are. J is strictly convex (its Hessian is $2(X^\top X + \lambda I)$), so the stationary point is the unique minimiser: $\hat{w}_\lambda = (X^\top X + \lambda I)^{-1}X^\top y$. ■

(b) If the columns of X are orthonormal ($X^\top X = I$), show that $\hat{w}_\lambda = \hat{w}_{OLS}/(1 + \lambda)$.

Answer

Here $\hat{w}_{OLS} = (X^\top X)^{-1}X^\top y = X^\top y$, and $\hat{w}_\lambda = (I + \lambda I)^{-1}X^\top y = \frac{1}{1+\lambda}X^\top y$. Every coefficient is shrunk by the same factor, and none becomes exactly zero (compare L_1).

M3.5 (*) A stronger penalty never fits the training data better**

For $\lambda \geq 0$ let $\hat{\theta}_\lambda \in \arg \min_\theta [\hat{R}(\theta) + \lambda\Omega(\theta)]$. Take $0 \leq \lambda_1 < \lambda_2$ and write $\hat{R}_k = \hat{R}(\hat{\theta}_{\lambda_k})$ and

$$\Omega_k = \Omega(\hat{\theta}_{\lambda_k}).$$

(a) Prove that $\Omega_2 \leq \Omega_1$ and $\hat{R}_1 \leq \hat{R}_2$.

Answer

By optimality of each solution for its own λ :

$$\hat{R}_1 + \lambda_1 \Omega_1 \leq \hat{R}_2 + \lambda_1 \Omega_2, \quad \hat{R}_2 + \lambda_2 \Omega_2 \leq \hat{R}_1 + \lambda_2 \Omega_1.$$

Adding them, the $\hat{R}$ terms cancel: $\lambda_1 \Omega_1 + \lambda_2 \Omega_2 \leq \lambda_1 \Omega_2 + \lambda_2 \Omega_1$, that is $(\lambda_2 - \lambda_1)(\Omega_2 - \Omega_1) \leq 0$. Since $\lambda_2 > \lambda_1$, $\Omega_2 \leq \Omega_1$. The first inequality then gives $\hat{R}_1 - \hat{R}_2 \leq \lambda_1(\Omega_2 - \Omega_1) \leq 0$. ■

(b) Interpret this result in terms of capacity.

Answer

Raising λ moves the solution to a smaller penalty budget and a worse training fit: a smaller effective family, with the parameter count unchanged. Whether the *validation* error improves depends on whether the lost fit was noise or signal, which is why λ is chosen on validation data.

M3.6 (**) The size of the winner's curse

Two configurations have the same risk R , and validation errors $\hat{R}_j = R + \varepsilon_j$ with $\varepsilon_1, \varepsilon_2$ independent $\mathcal{N}(0, \sigma^2)$.

(a) Show that $\mathbb{E}|W| = s\sqrt{2/\pi}$ for $W \sim \mathcal{N}(0, s^2)$.

Answer

By symmetry, $\mathbb{E}|W| = 2 \int_0^\infty w \frac{1}{s\sqrt{2\pi}} e^{-w^2/(2s^2)} dw$. With $u = w^2/(2s^2)$, $du = w dw/s^2$, this is $\frac{2s^2}{s\sqrt{2\pi}} \int_0^\infty e^{-u} du = \frac{2s}{\sqrt{2\pi}} = s\sqrt{2/\pi}$. ■

(b) Deduce that $\mathbb{E} \min(\hat{R}_1, \hat{R}_2) = R - \sigma/\sqrt{\pi}$.

Answer

Use $\min(a, b) = \frac{a+b-|a-b|}{2}$. Then $\mathbb{E} \min = R + \frac{\mathbb{E}\varepsilon_1 + \mathbb{E}\varepsilon_2}{2} - \frac{\mathbb{E}|\varepsilon_1 - \varepsilon_2|}{2}$. The difference is $\mathcal{N}(0, 2\sigma^2)$, so by (a) $\mathbb{E}|\varepsilon_1 - \varepsilon_2| = \sigma\sqrt{2} \cdot \sqrt{2/\pi} = 2\sigma/\sqrt{\pi}$, and $\mathbb{E} \min = R - \sigma/\sqrt{\pi}$. Neither model is better, yet the reported winner looks better. ■

M3.7 (**) What cross-validation actually estimates

N i.i.d. examples split into K equal folds. A fixed learning procedure trained on $n' = N(K-1)/K$ examples produces $\hat{f}_{n'}$.

(a) Show that $\mathbb{E}[\hat{R}_{CV}] = \mathbb{E}[R(\hat{f}_{n'})]$: the expected risk of the procedure at sample size n' , not at N .

Answer

For fold j , the model $\hat{f}^{(-j)}$ is trained on the other folds, which are n' i.i.d. examples, and validated on fold j , which is independent of them. Conditionally on the training folds the model is fixed, so (M2.7) $\mathbb{E}[\hat{R}_{\text{val}}^{(j)} | \mathcal{D}^{(-j)}] = R(\hat{f}^{(-j)})$. Taking expectations, $\mathbb{E}\hat{R}_{\text{val}}^{(j)} = \mathbb{E}R(\hat{f}_{n'})$, the same for every j . Averaging over the K folds keeps this value. ■

(b) Consequences: why is the estimate slightly pessimistic for the final model, and what happens when you report the minimum of $\hat{R}_{CV}$ over many configurations?

Answer

The final model is refitted on all $N > n'$ examples, and usually more data means lower risk, so the CV estimate is slightly too high for it. Reporting the minimum over many configurations reintroduces the winner's curse (M3.6): the selected CV score is optimistic. An untouched test set is still needed for the final number.

M3.8 (**) Precision depends on prevalence; F_1 is a harmonic mean

A detector has true-positive rate $t = \text{TPR}$ and false-positive rate $f = \text{FPR}$, independent of the prevalence π of positives.

(a) Show that precision $= \frac{\pi t}{\pi t + (1 - \pi)f}$, and that it increases with π when $t, f > 0$.

Answer

Among M cases there are πM positives, of which $\pi M t$ are flagged, and $(1 - \pi)M$ negatives, of which $(1 - \pi)M f$ are flagged. Precision is flagged positives over all flagged cases, so the M cancels. Write it as $(1 + \frac{1-\pi}{\pi} \cdot \frac{f}{t})^{-1}$: since $\frac{1-\pi}{\pi}$ decreases in π , precision increases. The same detector looks far worse on rarer attacks. ■

(b) Show that $F_1 = \frac{2TP}{2TP + FP + FN}$, that it does not depend on TN , and that $\min(P, R) \leq F_1 \leq \frac{P+R}{2}$.

Answer

With $P = \frac{TP}{TP+FP}$ and $R = \frac{TP}{TP+FN}$, we have $\frac{1}{F_1} = \frac{1}{2}(\frac{1}{P} + \frac{1}{R}) = \frac{(TP+FP)+(TP+FN)}{2TP}$, which gives the formula; TN never appears. A harmonic mean lies between min and max. It is at most the arithmetic mean because $\frac{P+R}{2} - \frac{2PR}{P+R} = \frac{(P-R)^2}{2(P+R)} \geq 0$. ■

4 Lecture 4: Discriminative and generative classification

Revision sheet: Lecture 4

The problem. The decision from a known posterior is easy; estimating $p(y | x)$ is the hard part. *Discriminative*: model $p(y | x)$ directly. *Generative*: model $\pi_c = p(y = c)$ and $p(x | y = c)$, then apply Bayes' rule.

Discriminative. Scores (logits) are turned into probabilities, then into an action. $\sigma(s) = \frac{1}{1+e^{-s}}$, $\text{softmax}_c(s) = \frac{e^{s_c}}{\sum_k e^{s_k}}$ (shift-invariant; for $C = 2$ it equals $\sigma(s_1 - s_0)$). Logistic regression: $s = w^\top x + b$, so the boundary is a hyperplane, and a cost threshold τ moves it to the parallel plane $s = \log \frac{\tau}{1-\tau}$. Empirical cross-entropy is $\frac{1}{N} \times$ the negative conditional log-likelihood. Deep classifiers learn a representation and end in scores. The discriminative route does not tell you whether x is familiar.

Generative. $p(x, c) = \pi_c p(x | c)$ and $p(c | x) = \frac{\pi_c p(x|c)}{\sum_k \pi_k p(x|k)}$, fitted by joint maximum likelihood. Two error sources: the wrong family (shape) and finite-sample estimation. Gaussian ML estimates: $\hat{\pi}_c = N_c/N$, $\hat{\mu}_c$ the class mean, $\hat{\Sigma}_c$ with denominator N_c .

LDA / QDA. LDA uses a shared Σ , giving linear scores $s_c = x^\top \Sigma^{-1} \mu_c - \frac{1}{2} \mu_c^\top \Sigma^{-1} \mu_c + \log \pi_c$ and a linear boundary. QDA uses its own Σ_c : $s_c = -\frac{1}{2} (x - \mu_c)^\top \Sigma_c^{-1} (x - \mu_c) - \frac{1}{2} \log |\Sigma_c| + \log \pi_c$, a quadratic boundary. A covariance has $d(d+1)/2$ free parameters. Shrinkage: $(1 - \lambda) \hat{\Sigma} + \lambda \alpha I$.

Naive Bayes. $p(x | c) = \prod_j p(x_j | c)$, score $\log \pi_c + \sum_j \log p(x_j | c)$. It is stable with little data; redundant features double-count evidence, giving overconfident probabilities. Variants: Gaussian (with variance smoothing), Bernoulli, Multinomial (with additive smoothing $\frac{n_{cj} + \alpha}{\sum_\ell n_{c\ell} + \alpha V}$), hybrid.

Imbalance, prior shift, calibration. Imbalance is an estimation issue. Prior shift ($p(x | y)$ fixed, π changes) is corrected generatively by swapping priors, and discriminatively by rescaling the odds: $r_d = \frac{(\pi_d/\pi_t)r_t}{(\pi_d/\pi_t)r_t + \frac{1-\pi_d}{1-\pi_t}(1-r_t)}$. Calibration is about the meaning of the probabilities; the threshold is about the action. **Familiarity**: a reject option, $\log p(x | c^*) < \tau$, rejects inputs unlike the training data. **Windows** aggregate flows over time (without leakage). **Language models**: generative pretraining, then a discriminative head.

4.1 Understanding questions

U4.1. “The decision is easy; the posterior is the problem.” Explain.

Answer

If $p(y | x)$ were known, Lecture 2 already gives the optimal action for any loss: the argmax, or a cost threshold. Everything difficult is in estimating the posterior from finite data, which is what the discriminative and generative routes are two ways of doing.

U4.2. What does “generative” name? Can a generative model classify?

Answer

It names the modelled direction: how inputs are produced given the class, $p(x | y)$, together with $p(y)$. It does not name the task. Yes, it classifies through Bayes' rule, and it could in principle also generate inputs of a chosen class.

U4.3. Does using a threshold of 0.5 make a classifier's boundary linear?

Answer

No. The boundary is the set where the score equals a constant, so its shape comes from the *score family*. Linear scores give hyperplanes; a nonlinear score $g(x)$ gives the curved set $g(x) = \text{const.}$

The threshold only selects *which* level set.

U4.4. Why is a maximum posterior of 0.99 not evidence that the input resembles the training data?

Answer

Softmax is a *relative* comparison among the known classes. An input far from all training data can still have one class score much larger than the others, which gives a posterior near 1. Familiarity is a statement about $p(x)$ or $p(x | c)$, which a discriminative model does not estimate. A generative likelihood check (a reject option) addresses it.

U4.5. Distinguish class imbalance, prior shift, the cost threshold and calibration.

Answer

Imbalance: few minority examples make the posterior hard to estimate; it is a data and estimation issue. *Prior shift*: the deployment class frequencies differ from training, so the posterior must be corrected. *Cost threshold*: given a correct posterior, it chooses the action according to error costs. *Calibration*: whether a stated probability u is right a fraction u of the time. Four separate questions, and they can all apply to the same model.

U4.6. The Gaussian densities in LDA are quadratic in x , yet LDA's boundary is linear. Why?

Answer

With a shared covariance, the quadratic term $-\frac{1}{2}x^\top \Sigma^{-1}x$ is identical in every class's log-density, so it cancels when two classes are compared. Only terms linear in x remain. "Linear" describes the boundary, not the densities.

U4.7. When would you prefer LDA to QDA? Use parameter counts in your answer.

Answer

QDA estimates one covariance per class, $C \cdot d(d+1)/2$ parameters (5050 per class at $d = 100$). LDA estimates one shared covariance. With few examples per class, QDA's covariances are unstable or singular (high variance), so LDA or shrinkage is preferred. With plenty of data and clearly different class shapes, QDA's flexibility pays off.

U4.8. Naive Bayes often classifies well but gives badly calibrated probabilities. Why?

Answer

The right argmax only needs the classes ranked correctly, which can survive a misspecified density. But when features are correlated within a class, the product of per-feature likelihoods counts the same evidence several times, which pushes the posterior toward 0 or 1. The decisions are often fine, while the probabilities are too extreme.

U4.9. Why is additive smoothing necessary in multinomial Naive Bayes? How do you handle a port value never seen in training?

Answer

A word with zero count in a class gives probability 0, and one unseen word zeroes the whole product, so no posterior can be computed. Smoothing adds α to every count, which keeps every probability positive and shrinks the estimates toward uniform. For an unseen port, create an "other" level during training, map unseen values to it at test time, and let smoothing give it nonzero probability.

U4.10. When would you choose a discriminative model, and when a generative one?

Answer

Discriminative: many labels, raw high-dimensional features that are hard to model, and accuracy as the goal; it spends no capacity on $p(x)$. Generative: few labels and a credible class model, where the structure reduces variance; when you need to handle missing features, check whether an input is compatible with known classes, or correct for prior shift by swapping priors. A wrong generative family creates bias that more data does not remove.

U4.11. Why can a per-flow classifier miss a port scan? What do window features add, and what are their pitfalls?

Answer

Each scan flow looks ordinary on its own, so every individual score stays low. A window aggregates recent behaviour from one source (flow count, distinct ports, failure fraction, score sum), which reveals the pattern. Pitfalls: a window shorter than the gap between flows sees too little; every feature must be computable at decision time (no future flows); a shared NAT address mixes several devices; and windows from the same episode must stay on one side of every split.

4.2 Quiz

- Q1.** Logistic regression with threshold 0.5 predicts class 1 exactly when
 (a) $s > 0.5$ (b) $s > 0$ (c) $s < 0$ (d) $\sigma(s) > 0$
- Q2.** Adding the same constant to every softmax score changes the probabilities. [True / False]
- Q3.** For $C = 2$, $\text{softmax}_1 = \sigma(s_1 - s_0)$. [True / False]
- Q4.** LDA assumes
 (a) a shared covariance (b) diagonal covariances (c) independent features (d) equal means
- Q5.** The number of free parameters in a $d \times d$ covariance matrix is
 (a) d^2 (b) $d(d+1)/2$ (c) d (d) $2d$
- Q6.** Naive Bayes assumes the features are marginally independent. [True / False]
- Q7.** Laplace smoothing ($\alpha = 1$) means
 (a) adding 1 to every count (b) multiplying the counts by 1 (c) removing rare words (d) normalising by V only
- Q8.** Under prior shift, the generative correction is to
 (a) refit the class models (b) swap the priors (c) change only the threshold (d) retrain from scratch
- Q9.** With a cost threshold, the logistic boundary moves to a parallel hyperplane. [True / False]
- Q10.** Minimising the empirical cross-entropy is maximising the
 (a) joint likelihood (b) conditional likelihood (c) prior (d) entropy
- Q11.** The ML covariance estimate divides by
 (a) N_c (b) $N_c - 1$ (c) N (d) d
- Q12.** QDA always beats LDA. [True / False]
- Q13.** Which route can naturally marginalise a missing feature?
 (a) discriminative (b) generative (c) neither (d) both equally

Answer key

- Q1 (b): $\sigma(s) > 0.5 \iff s > 0$.
 Q2 False: it is shift-invariant.
 Q3 True.
 Q4 (a).
 Q5 (b).
 Q6 False: conditionally independent *given the class*.
 Q7 (a).
 Q8 (b).
 Q9 True: $s = \log \frac{\tau}{1-\tau}$.
 Q10 (b).
 Q11 (a).
 Q12 False: it has more variance; with little data LDA often wins.
 Q13 (b).

4.3 Maths questions**M4.1 (★) Sigmoid identities**

$$\sigma(s) = 1/(1 + e^{-s}).$$

(a) Prove $\sigma(-s) = 1 - \sigma(s)$, $\sigma'(s) = \sigma(s)(1 - \sigma(s))$, and that the inverse of σ is the logit $\log \frac{p}{1-p}$.

Answer

$1 - \sigma(s) = \frac{e^{-s}}{1+e^{-s}} = \frac{1}{e^s+1} = \sigma(-s)$. Next, $\sigma'(s) = \frac{e^{-s}}{(1+e^{-s})^2} = \sigma(s) \cdot \frac{e^{-s}}{1+e^{-s}} = \sigma(s)(1 - \sigma(s))$. Finally, $p = \sigma(s) \iff \frac{p}{1-p} = \frac{\sigma(s)}{\sigma(-s)} = e^s \iff s = \log \frac{p}{1-p}$. So the score *is* the log-odds. ■

M4.2 (★★) Gradient and convexity of logistic regression

For one example, $\ell(w, b) = -y \log \sigma(s) - (1 - y) \log(1 - \sigma(s))$ with $s = w^\top x + b$ and $y \in \{0, 1\}$.

(a) Show that $\partial \ell / \partial s = \sigma(s) - y$, hence $\nabla_w \ell = (\sigma(s) - y)x$ and $\partial \ell / \partial b = \sigma(s) - y$.

Answer

By M4.1, $\frac{d}{ds} \log \sigma(s) = 1 - \sigma(s)$ and $\frac{d}{ds} \log(1 - \sigma(s)) = \frac{d}{ds} \log \sigma(-s) = -\sigma(s)$. So

$$\frac{\partial \ell}{\partial s} = -y(1 - \sigma) + (1 - y)\sigma = \sigma(s) - y.$$

The chain rule with $\partial s / \partial w = x$ and $\partial s / \partial b = 1$ finishes. The update is driven by the prediction error $\sigma - y$, scaled by the input. ■

(b) Show that the Hessian in w is $\sigma(s)(1 - \sigma(s))xx^\top$, which is positive semidefinite. Deduce that the empirical cross-entropy is convex in w .

Answer

Differentiate $(\sigma(s) - y)x$ once more: $\nabla_w^2 \ell = \sigma'(s)xx^\top = \sigma(1 - \sigma)xx^\top$. For any v , $v^\top(\sigma(1 - \sigma)xx^\top)v = \sigma(1 - \sigma)(v^\top x)^2 \geq 0$. A sum of convex functions is convex. So the training loss has no spurious local minima: every stationary point is a global minimum. ■

(c) For softmax with cross-entropy, $\ell = -\log p_y$ where $p = \text{softmax}(s)$, show that $\partial \ell / \partial s_c = p_c - \mathbf{1}(y = c)$.

Answer

$\ell = -s_y + \log \sum_k e^{s_k}$. The derivative of the first term is $-\mathbf{1}(y = c)$, and of the second $\frac{e^{s_c}}{\sum_k e^{s_k}} = p_c$. It is the same “prediction minus target” form as the binary case. ■

M4.3 (★) A generative posterior is a softmax of generative scores

Let $s_c(x) = \log \pi_c + \log p(x | c)$.

(a) Show that $p(c | x) = \text{softmax}_c(s(x))$, and that for two classes $p(1 | x) = \sigma(s_1(x) - s_0(x))$.

Answer

$\pi_c p(x | c) = e^{s_c(x)}$, so Bayes’ rule reads $p(c | x) = e^{s_c} / \sum_k e^{s_k}$. For $C = 2$, divide numerator and denominator by e^{s_1} : $p(1 | x) = \frac{1}{1 + e^{-(s_1 - s_0)}} = \sigma(s_1 - s_0)$. Both routes end in a softmax of scores; they differ in how the scores are obtained. ■

M4.4 (★★) LDA has exactly the logistic form

Two classes with $X | (Y = c) \sim \mathcal{N}(\mu_c, \Sigma)$ (shared Σ) and priors π_0, π_1 .

(a) Show that $\log \frac{p(1|x)}{p(0|x)} = w^\top x + b$ with $w = \Sigma^{-1}(\mu_1 - \mu_0)$ and $b = -\frac{1}{2}\mu_1^\top \Sigma^{-1}\mu_1 + \frac{1}{2}\mu_0^\top \Sigma^{-1}\mu_0 + \log \frac{\pi_1}{\pi_0}$.

Answer

The log-odds is $s_1 - s_0$ (M4.3). In each Gaussian log-density, the constants and $-\frac{1}{2} \log |\Sigma|$ are common to both classes and cancel. Expanding $-\frac{1}{2}(x - \mu_c)^\top \Sigma^{-1}(x - \mu_c) = -\frac{1}{2}x^\top \Sigma^{-1}x + x^\top \Sigma^{-1}\mu_c - \frac{1}{2}\mu_c^\top \Sigma^{-1}\mu_c$ (using the symmetry of Σ^{-1}), the $x^\top \Sigma^{-1}x$ terms cancel too. What remains is $x^\top \Sigma^{-1}(\mu_1 - \mu_0) - \frac{1}{2}\mu_1^\top \Sigma^{-1}\mu_1 + \frac{1}{2}\mu_0^\top \Sigma^{-1}\mu_0 + \log \frac{\pi_1}{\pi_0}$. ■

(b) Conclude that $p(1 | x) = \sigma(w^\top x + b)$. Why does this *not* make LDA and logistic regression the same method?

Answer

By M4.1, $p(1 | x) = \sigma(w^\top x + b)$. The posterior has the same *form*, but the parameters are found differently. LDA estimates μ_c, Σ, π by maximising the joint likelihood, which is efficient if the Gaussian assumption holds and biased if it does not. Logistic regression fits w, b directly by conditional likelihood, which makes no assumption on $p(x | y)$. With equal priors and $\Sigma = \sigma^2 I$ the boundary is the perpendicular bisector of the two means.

M4.5 (★★) QDA in one dimension

$X | (Y = c) \sim \mathcal{N}(\mu_c, \sigma_c^2)$ for $c = 0, 1$, with priors π_0, π_1 .

(a) Write the boundary equation $s_1(x) = s_0(x)$. Show that it is quadratic in x when $\sigma_0 \neq \sigma_1$, and linear when $\sigma_0 = \sigma_1$.

Answer

$s_c(x) = -\frac{(x - \mu_c)^2}{2\sigma_c^2} - \log \sigma_c + \log \pi_c$, up to a common constant. The boundary is

$$\left(\frac{1}{2\sigma_0^2} - \frac{1}{2\sigma_1^2}\right)x^2 + \left(\frac{\mu_1}{\sigma_1^2} - \frac{\mu_0}{\sigma_0^2}\right)x + \left(\frac{\mu_0^2}{2\sigma_0^2} - \frac{\mu_1^2}{2\sigma_1^2} + \log \frac{\sigma_0\pi_1}{\sigma_1\pi_0}\right) = 0.$$

The x^2 coefficient vanishes if and only if $\sigma_0 = \sigma_1$, which gives the LDA case. Otherwise there

can be two boundary points. The wider class then wins on *both* tails, because its density decays more slowly. ■

M4.6 (**) Bernoulli Naive Bayes is a linear classifier

Binary features $x_j \in \{0, 1\}$ with $p(x_j = 1 | c) = \theta_{cj} \in (0, 1)$, independent given the class.

(a) Show that the log-odds $\log \frac{p(1|x)}{p(0|x)} = \sum_j w_j x_j + b$, and give w_j and b .

Answer

$\log p(x_j | c) = x_j \log \theta_{cj} + (1 - x_j) \log(1 - \theta_{cj})$. Subtract class 0 from class 1, sum over j , and add the log prior ratio:

$$w_j = \log \frac{\theta_{1j}(1 - \theta_{0j})}{\theta_{0j}(1 - \theta_{1j})}, \quad b = \log \frac{\pi_1}{\pi_0} + \sum_j \log \frac{1 - \theta_{1j}}{1 - \theta_{0j}}.$$

So $p(1 | x) = \sigma(w^\top x + b)$: a linear boundary, with weights set by counting rather than by optimisation. ■

M4.7 (**) Duplicated features make Naive Bayes overconfident

A feature x_1 has log-likelihood ratio $\Lambda(x_1) = \log \frac{p(x_1|1)}{p(x_1|0)}$, and x_2 is an exact copy of x_1 .

(a) Show that the true log-odds is $\log \frac{\pi_1}{\pi_0} + \Lambda$, whereas Naive Bayes computes $\log \frac{\pi_1}{\pi_0} + 2\Lambda$. With m copies it computes $\log \frac{\pi_1}{\pi_0} + m\Lambda$.

Answer

Given x_1 , the copy carries no extra information: the true joint is $p(x_1, x_2 | c) = p(x_1 | c) \mathbf{1}(x_2 = x_1)$, and the indicator cancels in the ratio, leaving Λ . Naive Bayes multiplies $p(x_1 | c) p(x_2 | c)$ as if independent, so it adds Λ once per copy. ■

(b) Show that if $\Lambda > 0$ the Naive Bayes posterior exceeds the true one. Is the decision necessarily wrong?

Answer

σ is increasing and $\log \frac{\pi_1}{\pi_0} + 2\Lambda > \log \frac{\pi_1}{\pi_0} + \Lambda$ when $\Lambda > 0$, so $p_{NB}(1 | x) > p(1 | x)$: the probability is pushed toward the side the evidence favours. The decision can still be right whenever both log-odds have the same sign. The error is in the *calibration*, and it matters for cost thresholds, so removing redundant features improves the probabilities.

M4.8 (*) Additive smoothing shrinks toward uniform

Class c has token counts n_{cj} with total $n_c = \sum_j n_{cj}$, vocabulary size V , and $\hat{\theta}_{cj} = \frac{n_{cj} + \alpha}{n_c + \alpha V}$.

(a) Show that $\hat{\theta}_{cj} = \lambda \frac{n_{cj}}{n_c} + (1 - \lambda) \frac{1}{V}$ with $\lambda = \frac{n_c}{n_c + \alpha V}$, that the estimates sum to 1, and describe the limits $\alpha \rightarrow 0$ and $\alpha \rightarrow \infty$.

Answer

$\lambda \frac{n_{cj}}{n_c} + (1 - \lambda) \frac{1}{V} = \frac{n_{cj}}{n_c + \alpha V} + \frac{\alpha V}{n_c + \alpha V} \cdot \frac{1}{V} = \frac{n_{cj} + \alpha}{n_c + \alpha V}$. The sum over j is $\frac{n_c + \alpha V}{n_c + \alpha V} = 1$. As $\alpha \rightarrow 0$, $\lambda \rightarrow 1$ and we get the ML frequency n_{cj}/n_c , which can be 0. As $\alpha \rightarrow \infty$, $\lambda \rightarrow 0$ and we get the uniform $1/V$. More data (larger n_c) lets the counts dominate. ■

M4.9 (★★) Prior shift: the correction only rescales the odds

Training prior π_t , deployment prior π_d , the same $p(x | y)$, and a calibrated training posterior $r_t(x)$.

(a) Show that $\frac{r_d}{1-r_d} = \kappa \frac{r_t}{1-r_t}$ with $\kappa = \frac{\pi_d/(1-\pi_d)}{\pi_t/(1-\pi_t)}$, and derive the lecture's formula for r_d .

Answer

Bayes' rule in odds form: $\frac{r}{1-r} = \frac{\pi}{1-\pi} \cdot \frac{p(x|1)}{p(x|0)}$. The likelihood ratio is the same in training and deployment, so the two posterior odds differ only by the ratio of prior odds, κ . Solving $\frac{r_d}{1-r_d} = \kappa \frac{r_t}{1-r_t}$ for r_d gives $r_d = \frac{\kappa r_t}{\kappa r_t + 1 - r_t}$. Multiplying numerator and denominator by $\frac{1-\pi_d}{1-\pi_t}$ gives the lecture's form. ■

(b) Show that r_d is an increasing function of r_t . Deduce that thresholding r_d at τ is the same as thresholding r_t at some τ' , and find τ' .

Answer

Odds are increasing in r , and multiplying by $\kappa > 0$ preserves order, so r_d increases with r_t : the *ranking* of inputs is unchanged. Then $r_d > \tau \iff \frac{r_t}{1-r_t} > \frac{\tau}{(1-\tau)\kappa} \iff r_t > \tau'$ with $\frac{\tau'}{1-\tau'} = \frac{\tau}{(1-\tau)\kappa}$. For *decisions*, the prior-shift correction is a threshold change. For reported *probabilities* it is not, and it requires r_t to be calibrated.

M4.10 (★) Joint versus conditional likelihood

A generative model defines $p_\phi(x, y) = p_\phi(y | x) p_\phi(x)$.

(a) Show that $\log \prod_i p_\phi(x^{(i)}, y^{(i)}) = \sum_i \log p_\phi(y^{(i)} | x^{(i)}) + \sum_i \log p_\phi(x^{(i)})$, and interpret the second sum.

Answer

Take logs of the factorisation and sum over i . The first sum is the conditional log-likelihood, the only part a discriminative model optimises. The second rewards explaining the *inputs*, $p(x) = \sum_c \pi_c p(x | c)$. Maximising the joint spends modelling effort on $p(x)$. This helps when the model is right and data are few, and it can hurt the classification when the input model is wrong. ■

5 Numerical applications

The formulas from the four chapters, applied once to numbers. Logarithms are natural. All answers were checked by computer.

N1 (Lecture 2) Three actions

A flow has $p = P(\text{attack} \mid x) = 0.3$. Costs: false positive $C_{FP} = 5$, false negative $C_{FN} = 20$, human review $C_R = 3$ (after which the case is resolved correctly). Correct automatic decisions are free.

(a) Compute the three conditional risks and the optimal action. Give the decision regions in p .

Answer

Benign: $C_{FNP} = 6$. Attack: $C_{FP}(1 - p) = 3.5$. Review: 3. So review. Regions: benign for $p \leq C_R/C_{FN} = 0.15$, attack for $p \geq 1 - C_R/C_{FP} = 0.4$, review in between. Without review, the single threshold would be $5/25 = 0.2$.

N2 (Lecture 2) Three best predictions

$Y \mid x$ is exponential with mean 10: $F(y) = 1 - e^{-y/10}$.

(a) Give the Bayes prediction under squared loss, absolute loss, and the pinball loss with $c_u = 3$, $c_o = 1$.

Answer

Mean = 10. Median: $1 - e^{-a/10} = \frac{1}{2}$, so $a = 10 \ln 2 \approx 6.93$. Pinball level 3/4: $e^{-a/10} = \frac{1}{4}$, so $a = 10 \ln 4 \approx 13.86$.

N3 (Lecture 2) Entropy, cross-entropy, KL

$p = (0.6, 0.3, 0.1)$ is the true distribution and $q = (0.5, 0.4, 0.1)$ the model.

(a) Compute $H(p)$, $H(p, q)$ and $D_{KL}(p \parallel q)$, and check that $H(p, q) = H(p) + D_{KL}$.

Answer

$H(p) = 0.898$, $H(p, q) = 0.921$, $D_{KL} = 0.6 \ln \frac{0.6}{0.5} + 0.3 \ln \frac{0.3}{0.4} = 0.109 - 0.086 = 0.023$ nats. Indeed $0.898 + 0.023 = 0.921$.

N4 (Lecture 2) KNN before and after standardising

Training points as (income in k\$, age), with label: $A = (30, 25)$ class 0, $B = (32, 60)$ class 1, $C = (80, 30)$ class 0, $D = (85, 58)$ class 1, $E = (40, 40)$ class 1. Query $q = (35, 30)$. Use 3-NN with Euclidean distance.

(a) Predict with raw features.

Answer

Distances: A 7.07, E 11.18, B 30.15, C 45.0, D 57.3. The 3 nearest are A, E, B , with labels 0, 1, 1: predict **class 1**.

(b) Standardise each feature with the training mean and standard deviation ($1/N$ formula), then predict again.

Answer

The means are (53.4, 42.6) and the standard deviations (24.05, 14.25). The standardised distances are A 0.41, E 0.73, C 1.87, B 2.11, D 2.86. The 3 nearest are now A, E, C , with labels 0, 1, 0: predict **class 0**. Age has the smaller spread, so scaling gives age differences more weight relative to income: C (same age as the query) replaces B (30 years older), and the prediction flips.

N5 (Lecture 3) Ridge shrinkage

Data $(x, y) = (1, 2), (2, 3), (3, 5)$. A line through the origin, $f(x) = wx$, fitted with $\sum_i (y_i - wx_i)^2 + \lambda w^2$.

(a) Show that $\hat{w}_\lambda = \frac{\sum x_i y_i}{\sum x_i^2 + \lambda}$, then compute it for $\lambda = 0, 1, 10$.

Answer

Setting the derivative to zero, $-2 \sum x_i (y_i - wx_i) + 2\lambda w = 0$, gives $w(\sum x_i^2 + \lambda) = \sum x_i y_i$. Here $\sum x_i y_i = 23$ and $\sum x_i^2 = 14$, so $\hat{w} = 1.643, 1.533$ and 0.958 . The slope shrinks toward 0 as λ grows (M3.4 in one dimension).

N6 (Lecture 3) Choosing a threshold from a confusion matrix

1000 flows, 60 attacks. At threshold 0.5: $TP = 40, FP = 10, FN = 20, TN = 930$. At threshold 0.2: $TP = 54, FP = 60, FN = 6, TN = 880$.

(a) Compute accuracy, precision, recall, F_1 , FPR and balanced accuracy for both thresholds.

Answer

	Accuracy	Precision	Recall	F_1	FPR	BalAcc
0.5	0.970	0.800	0.667	0.727	0.011	0.828
0.2	0.934	0.474	0.900	0.621	0.064	0.918

(b) A missed attack costs 10 and a false alarm costs 1. Which threshold is better?

Answer

Cost at 0.5: $10 + 10 \times 20 = 210$. Cost at 0.2: $60 + 10 \times 6 = 120$. So 0.2 is better, even though accuracy and F_1 prefer 0.5. The metric must follow the costs.

N7 (Lecture 3) How noisy are the reported numbers?

(a) Two equally good configurations have validation noise $\sigma = 0.015$. By how much is the winner's reported error optimistic? And what is the standard error of a test accuracy of 0.92 measured on 2000 examples?

Answer

Optimism: $\sigma/\sqrt{\pi} = 0.015/1.772 = 0.0085$. Standard error: $\sqrt{0.92 \times 0.08/2000} = 0.0061$, so roughly 0.92 ± 0.012 at two standard errors.

N8 (Lecture 4) One logistic-regression step

$w = (2, -1)$, $b = -0.5$, $x = (1, 1)$, true label $y = 0$.

(a) Compute the score, the probability, the cross-entropy, and the gradient with respect to w . Then give the score boundary for a cost threshold $\tau = 0.3$.

Answer

The score is $s = 2 - 1 - 0.5 = 0.5$, so $\sigma(0.5) = 0.622$. The cross-entropy is $-\ln(1 - 0.622) = 0.974$, and the gradient is $(\sigma - y)x = (0.622, 0.622)$, so gradient descent lowers both weights. With $\tau = 0.3$, declare 1 iff $s > \ln \frac{0.3}{0.7} = -0.847$, that is $2x_1 - x_2 > -0.347$: a line parallel to the 0.5 boundary $2x_1 - x_2 = 0.5$.

N9 (Lecture 4) LDA in one dimension

$X | 0 \sim \mathcal{N}(0, 1)$ and $X | 1 \sim \mathcal{N}(3, 1)$, with $\pi_1 = 0.2$.

(a) Find the decision boundary and $p(1 | x = 2)$.

Answer

By M4.4 the log-odds is $3x - 4.5 + \ln \frac{0.2}{0.8}$. It is zero at $x^* = 1.5 + \frac{\ln 4}{3} = 1.962$, shifted from the midpoint 1.5 toward the rarer class. At $x = 2$ the log-odds is $6 - 4.5 - 1.386 = 0.114$, so $p(1 | 2) = \sigma(0.114) = 0.528$.

N10 (Lecture 4) Multinomial Naive Bayes, then prior shift

Vocabulary (offer, click, team, lunch). Spam counts (5, 4, 1, 0), ham counts (0, 1, 4, 5), $\alpha = 1$, priors spam 0.3 and ham 0.7. Message: “offer team team”, so $x = (1, 0, 2, 0)$.

(a) Compute the smoothed token probabilities and $p(\text{spam} | x)$.

Answer

Each class has 10 tokens, so the denominators are $10 + 4 = 14$. Spam: $(6, 5, 2, 1)/14$; ham: $(1, 2, 5, 6)/14$. The likelihoods are proportional to spam $0.3 \cdot 6 \cdot 2^2 = 7.2$ and ham $0.7 \cdot 1 \cdot 5^2 = 17.5$ (the common 14^3 cancels). So $p(\text{spam} | x) = 7.2/24.7 = 0.291$.

(b) A different, calibrated classifier was trained with $\pi_t = 0.5$ and outputs $r_t = 0.7$. In deployment $\pi_d = 0.1$. Correct the posterior.

Answer

$\pi_d/\pi_t = 0.2$ and $(1 - \pi_d)/(1 - \pi_t) = 1.8$, so $r_d = \frac{0.2 \times 0.7}{0.2 \times 0.7 + 1.8 \times 0.3} = \frac{0.14}{0.68} = 0.206$. The evidence is the same, but the base rate is lower, so the probability is lower.